NIDAR AirMouse — Component-by-Component Detailed Overview

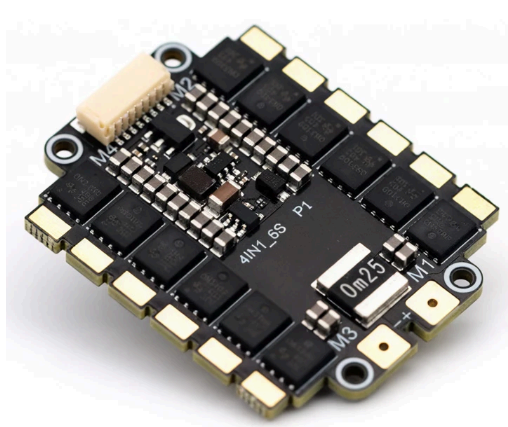
Propulsion System

T-Motor MN3510 (KV380) — Brushless Motor *(×4)*

An outrunner brushless motor: current pulses through stationary copper windings create a rotating magnetic field, which drags the outer magnet housing (bell) around with it — spinning the propeller directly with no gearbox. KV380 means it spins ~380 RPM per volt at no load, which on a 6S battery (22.2V) puts it in the right RPM range to drive a 12" prop efficiently. This is what actually generates lift.

TBS Lucid 6S 4-in-1 ESC (AM32 firmware)

Sits between the battery and the four motors. The flight controller sends a digital speed command (DShot) to the ESC; the ESC rapidly switches battery DC power through 3 phases per motor at the right timing to make it spin at the commanded speed. "4-in-1" means one board handles all four motors instead of four separate ESCs — less wiring, less weight. AM32 is open-source ESC firmware, giving bidirectional telemetry (RPM/voltage/current/temp per motor) back to the flight controller for logging and failure detection.



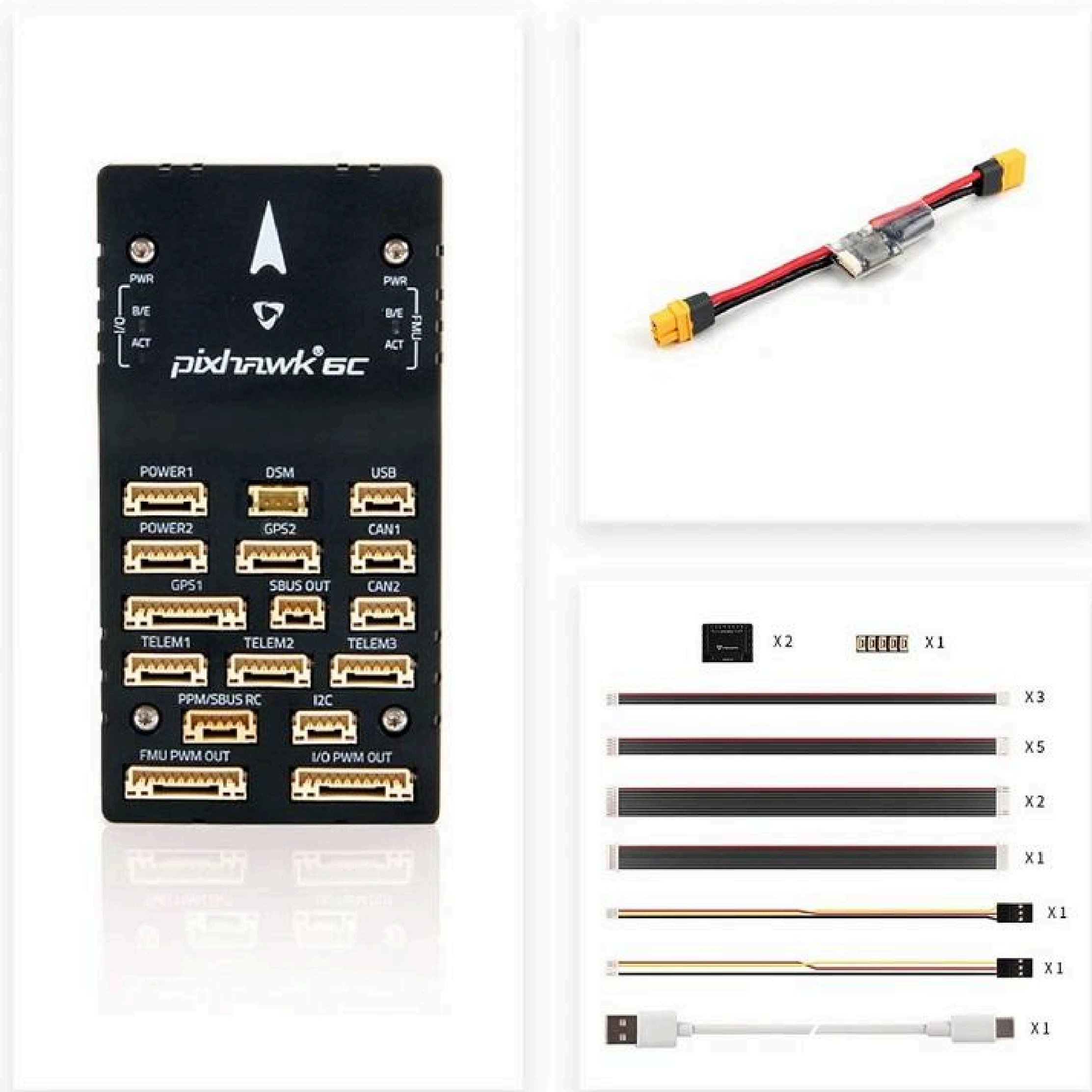
12×4 Carbon Fiber Propellers (2 CW + 2 CCW)

Convert the motor's spin into thrust by pushing air downward. Two spin clockwise, two counter-clockwise — this cancels out the reactive torque each motor would otherwise impart on the frame, which is why the quad doesn't spin itself. Carbon fiber keeps them light and stiff (less flex = more efficient thrust and less vibration feeding into the IMU).

6S LiPo Battery (4500–5000mAh, XT90)

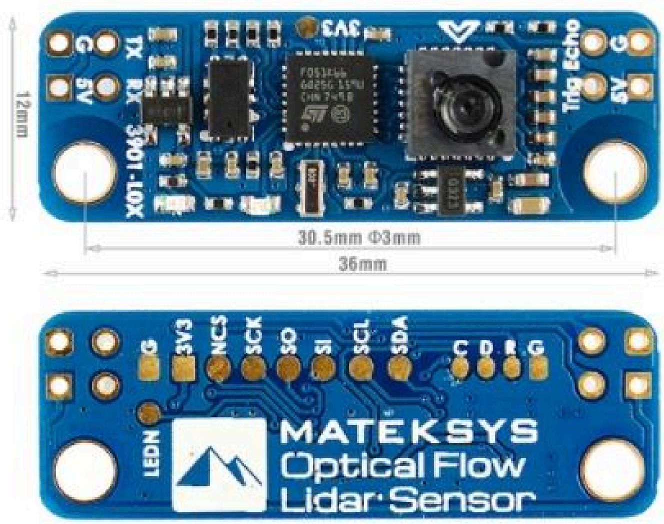
Six lithium-polymer cells wired in series to give 22.2V nominal (25.2V fully charged). This is the sole power source for the entire aircraft — motors, flight controller, and (through the UBEC) the Raspberry Pi and all sensors. The XT90 connector is rated for the high current draw four motors pull under load.

Flight Control System



Pixhawk 6C + PM02 — Flight Controller

The "brain" of flight stability. Contains an IMU (accelerometer + gyroscope) reading orientation hundreds of times a second, and runs the ArduPilot firmware's control loop: compare where the drone is pointed/moving vs. where it should be, and send corrected speed commands to each motor via the ESC — thousands of times per minute. The PM02 module sits between the battery and the flight controller, measuring voltage and current draw so the system knows remaining battery capacity in real time.



Matek 3901-L0X — Optical Flow + Rangefinder

This is the single most important sensor for a GPS-denied mission. It's two sensors on one board, mounted facing down:

- **PMW3901 optical flow chip** — essentially an optical-mouse sensor pointed at the ground. It captures ground-texture images at high speed and measures how the pattern shifts between frames, which the flight controller converts into horizontal velocity (how fast the drone is drifting sideways/forward).
- **VL53L0X laser time-of-flight sensor** — bounces an infrared laser pulse off the floor and times the return to measure precise height above ground.

Together, ArduPilot's EKF3 estimator fuses these two readings (instead of GPS) to hold position and altitude indoors — this is what lets the drone hover in place or navigate a room without any satellite signal.

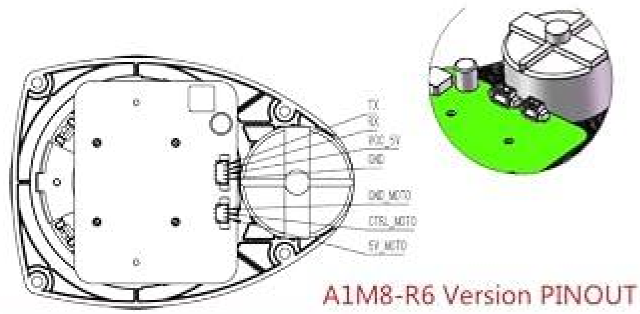
ExpressLRS Receiver — RC Link

Receives control stick inputs from the pilot's transmitter over 2.4GHz and passes them to the flight controller over the CRSF protocol. ExpressLRS is chosen here specifically for its low latency and strong performance in indoor, multipath-heavy (wall-reflecting) RF environments — critical since this is an indoor search mission, not open-field flying.

RFD900x (868MHz India variant) — Long-Range Telemetry Radio

A paired radio set: one on the drone, one on the ground station laptop. Continuously streams MAVLink telemetry (battery status, attitude, GPS-off/EKF health, mode) back to the operator, and serves as your safety-abort channel — if something goes wrong, this link is how you issue an emergency land/disarm command from the ground. The 868MHz variant is the legally required frequency band in India (915MHz is restricted there).


RPLIDAR



Interface	Signal Name	Type	Description	Min	Typical	Max
Motor Interface	5V_MOTO	Power	Power for RPLIDAR A1 Motor	-	5V	9V
	CTRL_MOTO	Input	Enable signal for RPLIDAR A1 Motor/PWM Control Signal	0V	-	5V_MOTO
	GND_MOTO	Power	GND for RPLIDAR A1 Motor	-	0V	-
Core Interface	VCC_5	Power	Power for RPLIDAR A1 Range Scanner Core	4.9V	5V	5.5V
	TX	Output	Serial output for Range Scanner Core	0V	-	5V
	RX	Input	Serial input for Range Scanner Core	0V	-	5V
	GND	Power	GND for RPLIDAR A1 Range Scanner Core	-	0V	V5.0

Figure 3-1 RPLIDAR A1 Pin Definition and Specification

RPLIDAR A2M8 — 2D LiDAR

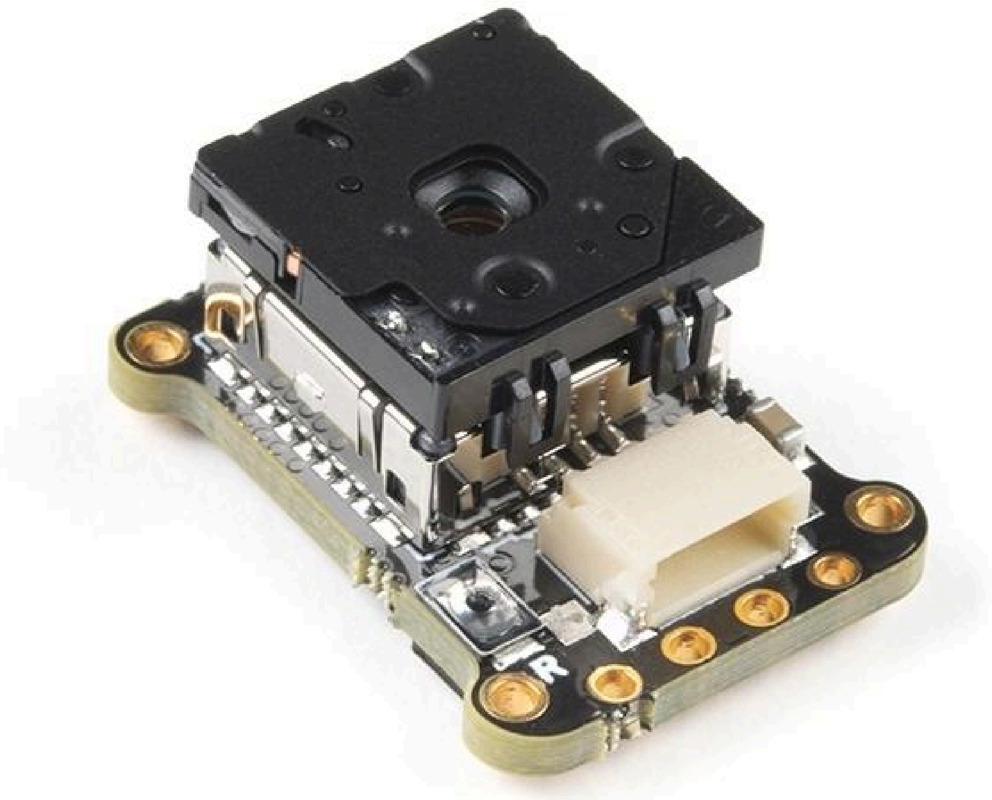
A spinning laser rangefinder that shoots out a rotating beam and measures the time it takes to bounce back off surfaces at every angle around a full 360°, at roughly 10 rotations per second. Each rotation produces a full ring of distance measurements — walls, doorframes, furniture, obstacles. As the drone moves, the SLAM software stitches thousands of these scans together into a 2D occupancy-grid map of the entire space. This is how the drone builds a floor plan of a building it's never seen before, with no prior map and no GPS.

Raspberry Pi 4 (8GB) — Companion Computer

The onboard "mission brain," separate from the Pixhawk. While the Pixhawk only worries about keeping the drone stable in the air, the Pi runs the higher-level software: SLAM mapping, camera/thermal processing, and the logic that decides where to fly next. It talks to the Pixhawk over MAVLink (via a serial/USB link) to send navigation commands and read back position/status. Needs active cooling because SLAM + vision + thermal processing running simultaneously is a sustained heavy CPU load.

Luxonis OAK-D — RGB + Depth Camera

Has two side-by-side cameras (like human eyes) that let it calculate depth by comparing the slight difference between the two images — this is stereo depth, giving distance-to-object without needing a separate laser sensor. It also has an onboard AI chip (Myriad X) that runs a person-detection neural network directly on the camera itself, so it can identify "there's a human-shaped object here" and report its distance — all without using any of the Raspberry Pi's own processing power.



FLIR Lepton 3.5 + PureThermal Mini Pro — Thermal Camera

Reads infrared radiation instead of visible light, giving an actual temperature value for every pixel in its image (this is called radiometric imaging). Anything warmer than its surroundings — most notably a human body — lights up clearly, even in darkness, smoke, or partial visual obstruction. Connects over USB as a standard video device (UVC), plus gives access to raw temperature data through its SDK. In this build it's cross-checked against the OAK-D's human-shape detection: thermal alone would flag any warm object (radiators, equipment), but thermal + visual shape confirmation together give a much more reliable survivor detection.

Powered USB3 Hub

The RPLIDAR, OAK-D, and thermal camera all draw more current and bandwidth combined than the Raspberry Pi's built-in USB ports can safely deliver. This hub has its own power input, so it feeds all three devices cleanly without starving the Pi's own power rail or causing random USB dropouts mid-flight — which would be catastrophic for SLAM/vision reliability.

Power Distribution

Dedicated 5V/5A UBEC

A separate voltage regulator stepping the battery's 22.2V down to 5V just for the Raspberry Pi and its sensors — kept electrically separate from the ESC's own power circuit. This matters because the LiDAR spinning up or the OAK-D running a burst of AI inference can cause a sudden current spike; without isolation, that spike could sag the voltage feeding the Pixhawk and destabilize the flight controller mid-flight.

How It All Connects